

Chapter 19 - Numerical Computation

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#Numerical Computation

Numerical computation is the science of using algorithmic, iterative processes and computer arithmetic to approximate solutions for complex mathematical problems, particularly when exact analytical solutions are impossible or impractical.

Characteristics of Numerical Computing

Accuracy	Accuracy refers to how well a method matches the expected result. It can also be interpreted as the degree to which a given quantity is correct and free from error.
Rate of convergence	Rate of convergence refers to how long it takes to apply a given method for a given time-step. If a sequence x_n converges linearly to l and there exists a number $\mu \in (0, 1)$ such that $\lim_{n \rightarrow \infty} \frac{ x_{n+1} - l }{ x_n - l } = \mu$ then the number μ is called the rate of convergence. 1. If $\mu = 0$ then the sequence is said to converge super-linearly. 2. If $\mu = 1$ then the sequence is said to converge sub-linearly.
Numerical stability	Numerical stability refers to how well the method copes with stiff constraints before error becomes unacceptably large.
Efficiency	Efficiency refers to the number of steps in the algorithm, the computer time, and the amount of the computer (or computing device) that is required.

Nonlinear equations

An equation not in the form of $y = ax + b$ where a and b are constants, is known as a non-linear equation.

#Root of an Equation

The value of the variable which satisfies the equation, is known as the root of an equation.

Graphically, the roots of an equation are the x-intercepts of its graph.

Some Useful Observations

1. The total number of roots of an algebraic equation is the same as its degree.
2. An algebraic equation can have at most as many positive roots as the number of changes of sign in $f(x)$.
3. An algebraic equation can have at most as many negative roots as the number of changes of sign in $f(-x)$.
4. If $f(x)$ is continuous in the interval $[a, b]$ and $f(a)f(b) < 0$ then a root must exist in the interval (a, b) .

Initial approximation

An initial approximation (x_0) is the starting guess for a root or solution in iterative numerical methods, such as Newton's method, chosen near the actual value to ensure convergence.

Convergence

Convergence ensures that an algorithm's approximate solution approaches the exact solution as iterations increase or grid spacing decreases.

If the method leads us close to the exact solution then we say the method is convergent. Otherwise, the method is called divergent.

Rate of convergence

The rate of convergence quantifies how quickly an iterative method or sequence approaches its limit (the true solution), typically measured by how the error decreases between iterations.

#Bisection Method

The bisection method is a root-finding algorithm which repeatedly bisects an interval then selects a sub-interval in which a root must lie for further processing.

Procedure to find the root of an equation by bisection method

- Let two initial points a and b such that:
 - $a < b$
 - $f(a)$ and $f(b)$ have opposite signs or $f(a)f(b) < 0$.
- Draw a table with headings:

a	b	$m = \frac{a+b}{2}$	$f(a)$	$f(b)$	$f(m)$	$b - a$
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- Fill the above values and check:
 - If $f(m) < 0$ then replace a by this m
 - If $f(m) > 0$ then replace b by this m
- Repeat the above process until you get the root within desired accuracy.
- If the pre-specified error is ϵ then the iterations should be done until $(b - a)$ becomes less than ϵ .

Number of iterations

When we have initial points a and b with a pre-specified error of ϵ then the number of iterations by which our approximation will have the provided error is

$$n \geq \frac{\log(b-a) - \log \epsilon}{\log 2}$$

#Newton-Raphson Method

The Newton-Raphson method is an efficient algorithm for finding successively better approximations to the zero (or roots) of a real-valued function.

Procedure to find the root of an equation by bisection method

- Let the initial guess be x_0 such that $f'(x_0) \neq 0$.
- Evaluate the successive guesses by the equation

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \text{ for } f'(x_n) \neq 0$$

Error after n^{th} iteration	Relative error = $\frac{ x_n - x_{n-1} }{x_n}$	Percentage error = $\frac{ x_n - x_{n-1} }{x_n} \times 100$
$e_n = x_n - x_{n-1} $		

Newton's iteration equation

To find the square root of a , let $f(x) = x^2 - a$ and the initial guess be x_0 such that $f'(x_0) \neq 0$ then

evaluate the successive guesses by the equation: $x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$ for $f'(x_n) \neq 0$.